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Plasma essential amino acid concentrations in response to casein infusion or ration change in dairy cows: A multilevel, mixed-effects meta-analysis

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ABSTRACT

The objectives of the current meta-analysis were to review the relationships between plasma individual essential AA concentrations and their respective AA digestible flows (AADI) in 2 independent sets of studies. The first set of studies included 36 casein infusion studies (CN; 83 treatment means) and was regarded as the best comparison standard available, because differences in AADI achieved by casein infusion (up to 40% of total metabolizable protein supply) did not rely on any model assumptions and were directly estimated from casein infusions. The second set of studies included 42 feeding trials (FT; 94 treatment means) in which AADI were predicted using the 2001 National Research Council model. The 2 sets of studies were not balanced for dry matter intake and the supplies of metabolizable protein and net energy for lactation; therefore, a subset of 17 CN studies (35 treatment means) and 19 FT trials (49 treatment means) balanced for these variables was assembled to allow the comparison of linear terms from CN and FT studies. In the subset of data set, the linear terms of individual AA did not differ between CN and FT studies except for Met and Thr, with a slope lower by 23 and 62%, respectively, in CN versus FT studies. The agreement in linear slopes between CN and FT studies indicates, indirectly, that AADI were predicted accurately by the National Research Council model. In the large data set, the relationships between plasma concentrations of Ile, Leu, Val, and their sum (branched-chain AA; BCAA) and their respective AADI shared common characteristics that distinguished them from the other AA. For the plasma concentration of BCAA, the linear terms were significant in CN and FT studies, but the quadratic terms were significant only in FT studies. This decline in the response of plasma concentration of BCAA to increased BCAA digestible flow in FT studies was associated with diets rich in energy, diets with a high concentrate level, or diets based on corn silage. These dietary conditions can stimulate insulin secretion and decrease plasma concentration of BCAA. For the non-BCAA, a quadratic term was significant for plasma His, Lys, Met, and Thr in each set of studies, indicating an increased removal of these AA by the liver as AADI increased.

Key words: dairy cow, meta-analysis, casein, amino acid

INTRODUCTION

The overall objective of new dairy cow-feeding programs is to increase the efficiency of N utilization. This implies balancing not only the supply with the requirement of MP, but also fine-tuning the ration for the supply of individual EAA to a particular production level. Indeed, recommendations are now made for Lys and Met in NRC (2001), for His, Lys, and Met in the Scandinavian Norfor (Volden, 2011), and for all EAA in INRA (INRA, 2018) and the Cornell Net Carbohydrate-Protein System (Van Amburgh et al., 2015). The efficacy of these dairy cow-feeding programs to adequately balance AA supplies with production targets will depend on their ability to accurately predict the digestible flows (**DI**) of individual AA. Besides direct comparison of predicted and measured EAA duodenal flows (e.g., Pacheco et al., 2012), plasma concentration of EAA might be a valuable surrogate to estimate the accuracy of predicted EAA supply.

However, a primary challenge in studying peripheral plasma EAA concentrations is that several mechanisms are operating in coordination at the organ and interorgan levels to maintain AA homeostasis, and these mechanisms differ between EAA. The liver extracts substantial proportions of His, Met, Phe, and Trp hepatic inflows, whereas the liver removes very little Lys and branched-chain AA (BCAA; Hanigan, 2005; Lapierre et al., 2005). Although the hepatic removal of BCAA is regulated by the hormonal status (Brockman

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et al., 1975), extra-hepatic tissues also have a major role in BCAA utilization and catabolism, which is exacerbated during an acute insulin challenge (Brockman et al., 1975) and under the hyperinsulinemic-euglycemic clamp procedure (Laarveld et al., 1981).

At the mammary gland level, the uptake of Lys and the BCAA exceeds their secretion into milk protein and this excess increases with increased supply (Lapierre et al., 2012). All these interorgan processes may cause asynchronous regulation of the difference (**Δ**) in plasma concentration of individual AA (**p[AA]**). Nevertheless, a very close relationship exists between incremental AADI and p[AA]; for example, with graded omasal Lys infusions (Borucki-Castro et al., 2008a). Similarly, in casein infusion (**CN**) studies, ΔAADI is accurately and experimentally induced under constant dietary conditions in the casein-infused and control treatments. Therefore, the best comparison standard available for responses in p[AA] to ΔAADI consists of relationships from CN studies, because ΔAADI cannot be estimated with greater accuracy in other types of studies. In feeding trials (**FT**), ΔAADI corresponds to the difference between the highest and lowest (usually the control treatment) AADI predicted by the selected dairy cow feeding program, which includes aggregated sources of error (Pacheco et al., 2012). The response in p[AA] to incremental duodenal flow of AA (**AADuo**) was reviewed by Patton et al. (2015) using a large data set containing mainly FT studies and few AA or CN infusion studies. However, the corresponding relationships for CN studies have not been yet evaluated to our knowledge. Martineau et al. (2017) reviewed the response of p[AA] to incremental MP but not to incremental AADuo in CN studies.

The current study was undertaken because notable differences were observed between results reported in Patton et al. (2015) and responses in p[AA] to incremental AADuo in CN studies using the data set used by Martineau et al. (2017; refer to Supplemental File S1 and Table S1 for additional information, https://doi.org/10.3168/jds.2018-15218). For example, the linear terms of Ile, Leu, and Val were about 2-fold lower in FT versus CN studies; this indicated that, for a same ΔBCAADuo, Δp[BCAA] in FT studies was half that in CN studies. This difference between experimental approaches was unexpected and suggested an overestimation of BCAADuo by the feeding program used in Patton et al. (2015). Underestimations of BCAADuo by feeding programs were more likely to occur because of the incomplete recovery of free AA after the usual 24-h hydrolysis of proteins for BCAA (Rowan et al., 1992). Indeed, the results from the 2 sets of studies could not be directly compared because AADuo were not predicted using the same feeding program and the 2 sets of studies were not balanced for MP supply (25% less MP supply in CN vs. FT studies). Therefore, a new analysis was required in which AADI (not AADuo) would be predicted using the same feeding program, and the comparison of relationships would be conducted on a subset of CN and FT studies balanced for DMI and the supplies of MP and NE_L.

Our hypotheses were that (1) the linear slopes would be similar in the subset of CN and FT studies balanced for DMI and the supplies of MP and $NE_{L,}$ assuming that predictions of AADI by NRC (2001) are adequate, and (2) the relationships between p[AA] and AADI would differ between BCAA and the other EAA in the data set with all the studies. The relationships would include only a linear component for the BCAA (i.e., very little liver removal of BCAA) and a quadratic component for the other EAA (i.e., significant liver removal of EAA). Therefore, the objectives of the current meta-analysis were to (1) test whether the terms relating p[AA] and AADI differ between CN and FT studies in a subset of data balanced for DMI and the supplies of MP and NE_L, (2) review the relationships between p[AA] and AADI in CN and FT studies in the large data set, and (3) explore the influence of moderators on the relationships between p[AA] and AADI in FT studies, but not in CN studies, because moderators related to the diet were constant on a within-study basis in CN studies.

MATERIALS AND METHODS

Data Set

A large collection of studies reporting p[AA] and milk true protein yield (**MTPY**) in dairy cows was assembled by merging 141 treatment means from Martineau et al. (2014, 2017) and 526 treatment means from Patton et al. (2015). Data from Schwab et al. (1992) and Larsen et al. (2014) were removed from the data set because control cows were infused with DL-Met, L-Lys, or both and cows were in early postpartum period, respectively. Noncasein infusion and RPAA studies (185 treatment means) were removed from the data set, but not data from studies with postruminal infusion of a casein-like mixture of AA (e.g., Schei et al., 2007a,b). The data set was updated in June 2017 using the Scopus online database to include 40 new treatment means from dairy cow studies published since 2012 in the *Journal of Dairy Science*. Note that the data sets from Martineau et al. (2014, 2017) were updated on regular basis by the first author to include canola meal feeding and casein infusion studies with no restraint to the *Journal of Dairy Science*.

The expanded data set included 83 and 450 treatment means from CN and FT studies, respectively. The supply of MP did not vary in several FT studies, but it did in all CN studies by study design. To narrow the number of treatment means from FT studies, it was decided a priori to remove from the data set all FT studies with within-study coefficients of variation for ΔMP supply <0.05 (cutoff value in CN studies). Therefore, the final data set included 36 CN studies (83 treatment means) and 42 FT trials (94 treatment means; Table 1). The final data set was divided to include CN and FT studies balanced for DMI and the supplies of MP and NE_L (Table 1; Figure 1). The subset of data set included 17 CN studies (35 treatment means) and 19 FT trials (49 treatment means). The summary statistics for cow-associated and moderator variables are reported in Table 2 and 3, respectively.

The following terminology will be used throughout this manuscript: a publication could report more than a single experiment (e.g., Choung and Chamberlain, 1995a), each experiment could report also more than 1 study (e.g., Acharya et al., 2015), and each study included at least 2 treatment means. Note that p[AA] was assumed to be in micromoles per deciliter (not μmol/mL) in Seymour et al. (1990) and in micromolar (not mmol/L) in Choung and Chamberlain (1993b).

Calculations

All diet characteristics were calculated using NRC (2001), as outlined in Martineau et al. (2011). To compute MP supplied by casein, CP was converted into MP assuming (1) 91.0% DM, (2) 92.7% CP (DM basis; NRC, 2001), and (3) 100% intestinal CP digestibility (Huhtanen and Hristov, 2010). For all studies, the AADI from casein was calculated based on the profile of sodium caseinate (Galindo et al., 2011). For example, the infusion of 1.0 kg of casein (as-is basis) would yield 843.6 g of MP and 82.1 g of LeuDI.

To compute NE_L (Mcal/d) supplied by casein, ME was converted into NE_L using equation 2–11 from NRC (2001), assuming 5.08 Mcal of ME/kg of DM (NRC, 2001). Similar assumptions were made for any casein-like AA mixture (Schei et al., 2007a,b). Balances in MP (kg/d) and NE_L (Mcal/d) were the estimated supply minus requirements for maintenance and lactation (NRC, 2001). If not reported, MTPY was 0.95 × milk CP yield (Lapierre et al., 2012).

Outcomes were weighted by the inverse of the squared standard error in the statistical model. If not reported, standard error was computed by dividing standard deviation and standard error of the difference between any 2 means by $\sqrt{n}$ and $\sqrt{2}$, respectively (n being the number of replicates; Steel et al., 1997). Reported p[BCAA] was used when available (70 treatment means), otherwise p[BCAA] was calculated as the sum of p[Ile], p[Leu], and p[Val]. Unless reported, standard error of p[BCAA] was estimated by propagation of error (Neter et al., 1988) and a correction was applied due to the correlation among Ile, Leu, and Val. Briefly, studies simultaneously reporting p[BCAA], p[Ile], p[Leu], and p[Val] were used to compute the ratio between standard deviations of BCAA and propagated standard deviations of Ile + Leu + Val. The ratio was 1.56, indicating that the correlation among p[Ile], p[Leu], and p[Val] had to be taken into account (see Martineau et al., 2017 for more details). Therefore, standard error of p[BCAA] was computed by dividing (1.56 × propagated $SD_{Ile+Leu+Val}$) by $\sqrt{n}$ in publications not reporting p[BCAA] but reporting p[Ile], p[Leu], and p[Val].

Outcomes and Moderators

The outcomes were MTPY (g/d), p[urea] (m*M*), and p[AA] (μ*M*). The moderator of interest was MP supply (kg/d; diet plus infusate) for MTPY and p[urea], and AADI (g/d; diet plus infusate) for the respective p[AA]. Eight continuous moderators were calculated: total NE_L supply (Mcal/d) and dietary NE_L concentration (Mcal/kg), balances of MP (kg/d) and NE_L (Mcal/d), dietary proportion of concentrate (% DM basis), dietary concentrations of CP (% of diet DM), RUP or RDP (% of CP), NDF, and NFC (% of diet DM; Table 3). The correlation coefficients between moderators were calculated using the cor.test function from the statistics package in R (Table 4). Continuous moderators were categorized into 2 groups to perform additional analyzes on a between-study basis. All treatment means from the same study were designated as negative or positive MP and NE_L balance, and as low or high level of NE_L supply and concentration, concentrate, CP, RUP, RDP, NDF, and NFC based on the mean of each moderator within the study compared with the overall mean in FT studies (Table 3).

Three categorical moderators were evaluated in the meta-regressions: breed, forage, and stage of lactation. Breed was dichotomized as 0 = Ayrshire (including also Norwegian, Swedish, and Nordic red cows) and 1 = Holstein. Forage was dichotomized as 0 = grass/legume and 1 = mixed if the proportion of corn silage in the forage component of the diet was >0.25 (DM basis). Stage of lactation was dichotomized as 0 = early (≤100 d) and 1 = mid–late (>100 d). Days in milk were not reported in 5 publications; therefore, stage of lactation was (1) as reported in the publication for Derrig et al.

(1974) and Clark et al. (1977) or (2) based on the average MTPY in the data set (early >847 g/d; mid–late ≤847 g/d) for Spires et al. (1975), Rogers et al. (1984), and Choi et al. (2013). Means among categorical moderators and sets of categorical data were compared by one-way ANOVA and Pearson's chi-squared tests, respectively, using the anova and chisq.test functions from the statistics package in R.

Table 1. Characteristics of publications[1]

Authors and year of publication[2]	Type of study	Details if needed	Number: Studies	Number: Treatments
Acharya et al. (2015)	Feeding trial	—	2	4
Arriola Apelo et al. (2014)	Feeding trial	Positive and negative control	1	2
Bach et al. (2000)	Feeding trial	Low-quality AA profile only	1	2
# Blouin et al. (2002)	Feeding trial	—	1	2
Borucki Castro et al. (2008b)	Feeding trial	—	3	4
Cabrita et al. (2011)	Feeding trial	Soybean meal only	1	2
# Cant et al. (1993)	Casein infusion	—	2	4
# Carder and Weiss (2017)	Feeding trial	—	1	2
# Casper et al. (1990)	Feeding trial	—	2	4
Choi et al. (2013)	Casein infusion	Control and casein	1	2
Choung and Chamberlain (1992a)	Casein infusion	Basal and casein	1	2
Choung and Chamberlain (1992b)	Casein infusion	Experiment 1; basal and casein	1	2
Choung and Chamberlain (1993a)	Casein infusion	Casein and soybean protein isolate	2	7
Choung and Chamberlain (1993b)	Casein infusion	—	3	6
Choung and Chamberlain (1995a)	Casein infusion	Experiment 1; basal and caseinate	1	4
Choung and Chamberlain (1995a)	Casein infusion	Experiment 2; basal and caseinate	1	3
Choung and Chamberlain (1995b)	Casein infusion	Experiment 2; basal and caseinate	1	2
Christensen et al. (1994)	Feeding trial	14.2 and 17.5% CP	1	2
# Clark et al. (1977)	Casein infusion	—	2	4
Davidson et al. (2003)	Feeding trial	Urea treatment not included	2	3
Derrig et al. (1974)	Casein infusion	—	1	2
Garnsworthy et al. (2008)	Feeding trial	Low Leu treatments	1	2
Giallongo et al. (2015)	Feeding trial	—	1	3
Giallongo et al. (2016)	Feeding trial	—	1	2
# Gidlund et al. (2015)	Feeding trial	—	2	7
# Griinari et al. (1997)	Casein infusion	No insulin	1	2
Guinard and Rulquin (1994)	Casein infusion	—	1	4
# Hanigan et al. (2004)	Casein infusion	—	1	4
Hu et al. (2007)	Feeding trial	DCAD = 22 and 47 mEq/100 g of DM	2	4
Kleinschmit et al. (2007)	Feeding trial	Mixed forage treatment excluded	1	2
# Korhonen et al. (2002)	Feeding trial	Corn gluten meal excluded	2	3
# Kung et al. (1984)	Feeding trial	11.3 and 14.6% CP (soybean meal)	1	2
Lee et al. (2015)	Feeding trial	—	1	2
Mackle et al. (1999)	Casein infusion	—	2	4
Maxin et al. (2013)	Feeding trial	Corn distillers grain excluded	1	3
# Miettinen and Huhtanen (1997)	Casein infusion	—	2	4
Piepenbrink et al. (1996)	Feeding trial	—	1	2
# Puhakka et al. (2016)	Feeding trial	—	2	6
# Rae et al. (1983)	Feeding trial	—	2	4
Raggio et al. (2004)	Feeding trial	Medium and high MP	1	2
# Raggio et al. (2006)	Casein infusion	—	2	4
# Rinne et al. (1999)	Feeding trial	—	2	7
# Rinne et al. (2015)	Feeding trial	—	2	5
Rius et al. (2010a)	Casein infusion	Starch and starch + casein	1	2
Rius et al. (2010b)	Feeding trial	—	2	4
# Rogers et al. (1984)	Casein infusion	Water and caseinate	1	2
# Schei et al. (2007a)	Casein infusion	Abomasal infusion (early lactation)	1	2
Schei et al. (2007b)	Casein infusion	Abomasal infusion (late lactation)	1	2
# Seymour et al. (1990)	Casein infusion	Soybean meal only	2	3
# Shingfield et al. (2003)	Feeding trial	—	2	7
# Spires et al. (1975)	Casein infusion	—	1	2
# Vanhatalo et al. (2003a)	Casein infusion	—	2	4
Vanhatalo et al. (2003b)	Casein infusion	Control and casein	1	2
Whitelaw et al. (1986)	Casein infusion	—	1	4

[1]Studies marked with a hash (#) symbol were used to compare linear terms from linear relationships in casein infusion studies and feeding trials (refer to Table 5).

Multilevel Mixed-Effects Meta-Regression Model

Three analyses were conducted: (1) the comparison of linear terms between CN and FT studies in a subset of data set that included CN and FT studies balanced for DMI and the supplies of MP and NE_L (Table 5 and Figure 1); (2) fitting linear and quadratic models to CN and FT studies in the large data set (Table 6 and Figures 2, 3, and 4); and (3) investigating the influence of moderators on the linear relationships in FT studies

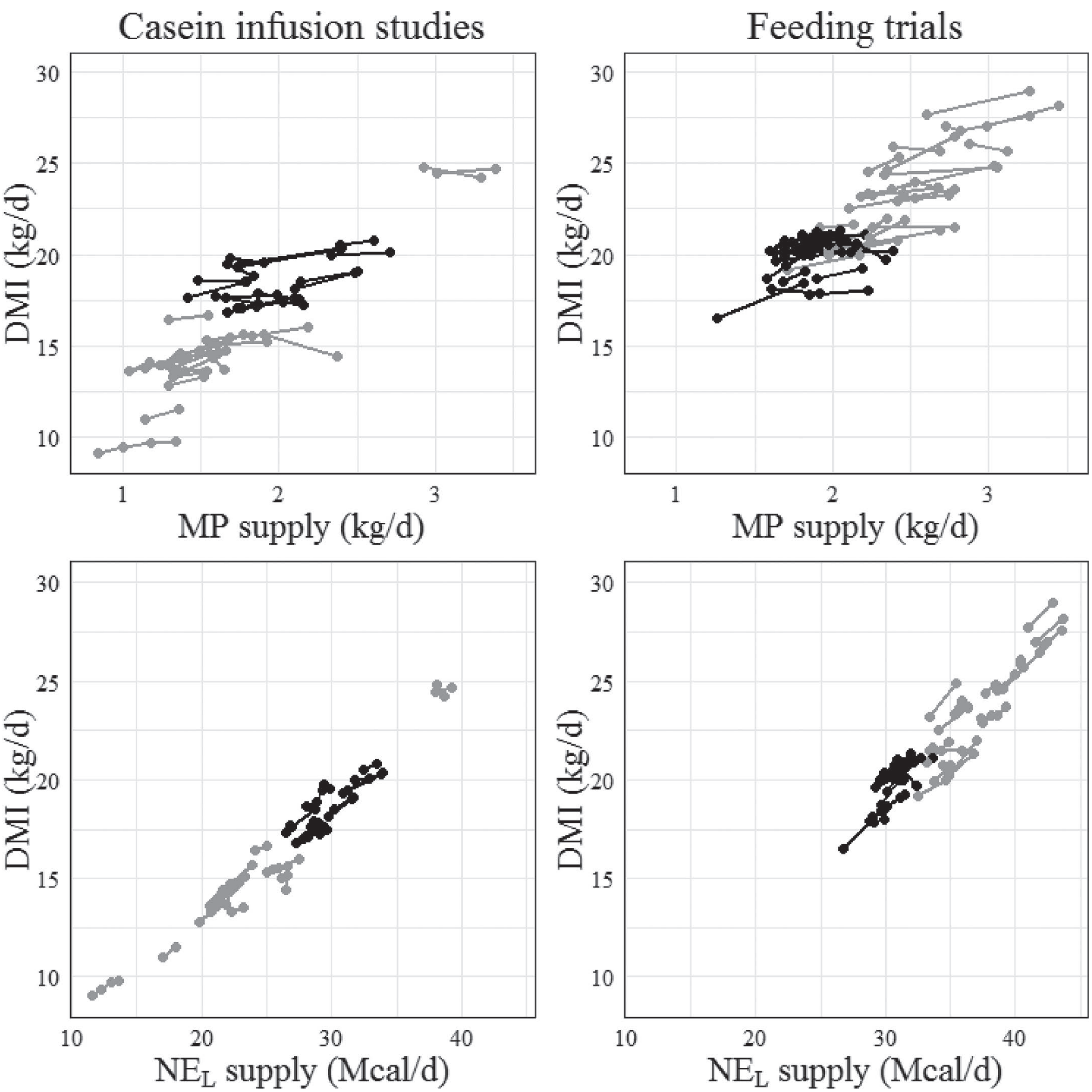

Figure 1. Dry matter intake (kg/d) versus MP supply (kg/d; upper panels) and NE_L supply (Mcal/d; bottom panels) in casein infusion studies and feeding trials. Studies in black were used to compare relationships in casein infusion studies and feeding trials. Each study is depicted by a connected line.

from the large data set (Table 7). In each analysis, data were fitted to a 3-level mixed-effects meta-regression model (each observation or line nested within a study itself nested within an experiment) using the rma.mv function in the metafor package in R. In the data set, 10 control observations were duplicated (2 CN and 8 FT); for example, 3 separate studies (1 for each source of soybean meal) were created after duplicating the negative control twice in Borucki Castro et al. (2008b).

The sampling variance (**vi**) corresponding to each outcome was specified as the squared standard error (Viechtbauer, 2010, 2016a,b). The weight matrix was the inverse of the marginal variance-covariance matrix, which is block-diagonal with $vi + \sigma 2_{experiment} + \sigma 2_{study}$ along the diagonal and $\sigma 2_{experiment}$ for off-diagonal elements that correspond to the same level of experiment. In these models, the weight matrix ignored the underlying but unknown correlation of sampling errors within clusters. Therefore, unbiased estimates of fixed effects and valid estimates of standard error were obtained using the robust function in the metafor package with the outermost nesting variable (i.e., experiment) as the clustering variable. Use of the robust function does not change the weight matrix, but only affects the way the variance-covariance matrix of the fixed effects and downstream standard error and *P*-values are computed (Viechtbauer, 2017a,b,c).

In the first analysis, the meta-regressions were conducted in CN and FT studies separately. The comparison of average effect size estimates from CN and FT studies was done using all studies and allowing the amount of residual heterogeneity to be different in each set of studies (Viechtbauer, 2017d). For this purpose, a new dichotomous moderator was created to distinguish the 2 sets of studies (CNFT; 0 = CN and 1 = FT) and the statistical model included a random argument with the following inner and outer formula: random = list(~factor(CNFT)|experiment, ~factor(CNFT)|line). The variance structure was specified as struct = c("DIAG,""DIAG") (Pustejovsky, 2017). In the second analysis, linear and quadratic relationships were evaluated in CN and FT studies separately. In the third analysis, the influence of each moderator on linear relationships was tested only in FT studies because moderators related to the diet were constant on a within-study basis in CN studies (refer to Supplemental Figure S1; https://doi.org/10.3168/jds.2018-15218). More details on the statistical model can be found in Martineau et al. (2017).

The generalized or weighted least squares extension of Cochran's *Q*-test when moderators are included in the model (**Q_E-test**) was computed for each model. This procedure tests whether the variability in the observed outcomes that was not accounted for by the

Table 2. Summary statistics for the cow-associated variables in casein infusion studies and feeding trials

Variable[1]	Type of study	N[2]	Mean	SD	Minimum	Maximum
Milk true protein yield (g/d)	Casein infusion	83	710	188.9	317	1,089
	Feeding trial	94	970	146.5	620	1,425
Plasma urea (m*M*)	Casein infusion	51	4.45	1.476	1.98	7.43
	Feeding trial	57	4.06	1.329	1.19	6.49
Plasma concentration (μ*M*)						
Arginine	Casein infusion	83	77.1	16.74	39.0	127.2
	Feeding trial	83	84.9	25.63	47.4	153.0
Histidine	Casein infusion	83	40.8	15.72	9.0	85.0
	Feeding trial	90	44.3	15.22	12.1	94.6
Isoleucine	Casein infusion	83	115.8	30.22	61.7	193.0
	Feeding trial	94	128.5	26.40	72.9	187.0
Leucine	Casein infusion	83	128.6	48.40	60.1	258.4
	Feeding trial	94	151.1	42.26	68.9	260.0
Lysine	Casein infusion	83	83.4	23.23	20.0	133.0
	Feeding trial	94	84.8	15.77	42.0	125.0
Methionine	Casein infusion	81	20.7	5.43	8.0	37.1
	Feeding trial	94	20.4	6.02	9.3	40.5
Phenylalanine	Casein infusion	83	47.4	6.90	35.0	77.0
	Feeding trial	94	48.2	9.51	27.2	96.0
Threonine	Casein infusion	83	108.8	23.68	54.0	163.0
	Feeding trial	94	105.2	18.40	74.0	162.5
Valine	Casein infusion	83	229.3	76.97	111.8	439.0
	Feeding trial	94	258.5	63.53	139.0	452.0
Branched-chain AA	Casein infusion	83	474.0	149.37	248.8	841.9
	Feeding trial	94	538.0	119.42	290.0	853.5

[1]Branched-chain AA = Ile + Leu + Val.

[2]Number of observations.

moderators included in the model was larger than one would expect based on the sampling variability and the given covariances among the sampling errors alone (Viechtbauer, 2018). The higher the Q_E-test value, the greater the unexplained variance left between all observed outcomes in the data set after moderators have been added to the meta-analytic model.

Potential outlying and influential observations and studies were detected using the rstandard, the rstudent, and the cook.distance functions in the metafor package. All relationships were graphed using the ggplot function of the ggplot2 package for R (Wickham, 2009) to evaluate the relationships. Other functions in the metafor package were used to reveal (1) profile likelihood plots of the variance components (profile.rma.mv), (2) various aspects and assumptions of the data (qqnorm), and (3) the occurrence of a publication bias (funnel).

Probabilities ≤ 0.05 will be referred to as significant and probabilities of $0.05 < P \leq 0.10$ will be referred to as a tendency in the manuscript. The experimental protocol was approved by the Institutional Animal Care Committee of the Sherbrooke Research and Development Centre according to the Canadian Council on Animal Care (2009).

Table 3. Summary statistics for the moderator variables in casein infusion studies and feeding trials

Variable[1]	Type of study	N[2]	Mean	SD	Minimum	Maximum
DMI (kg/d)	Casein infusion	83	16.3	3.28	9.1	24.8
	Feeding trial	94	21.8	2.64	16.5	29.0
MP supply (kg/d)	Casein infusion	83	1.77	0.506	0.84	3.38
	Feeding trial	94	2.20	0.439	1.26	3.44
NE_L supply (Mcal/d)	Casein infusion	83	25.8	5.65	11.6	39.2
	Feeding trial	94	34.0	4.10	26.7	43.7
NE_L concentration[3] (Mcal/kg)	Casein infusion	83	1.55	0.090	1.27	1.74
	Feeding trial	94	1.56	0.067	1.42	1.73
MP balance (kg/d)	Casein infusion	83	0.10	0.290	−0.53	1.06
	Feeding trial	94	−0.08	0.304	−0.64	0.78
NE_L balance (Mcal/d)	Casein infusion	83	−0.34	3.168	−7.71	7.96
	Feeding trial	94	0.53	3.582	−6.66	9.56
Concentrate[3] (%, DM basis)	Casein infusion	83	40.8	11.74	21.6	57.8
	Feeding trial	94	47.5	7.10	35.1	70.0
CP[3] (% of DM)	Casein infusion	83	15.7	1.93	13.4	23.3
	Feeding trial	94	16.5	1.80	11.3	20.9
RUP[3] (% of CP)	Casein infusion	83	33.7	7.86	20.1	50.4
	Feeding trial	94	34.8	5.66	22.7	55.5
NDF[3] (% of DM)	Casein infusion	83	39.1	7.05	24.8	57.1
	Feeding trial	94	36.6	6.08	25.0	48.9
NFC[3] (% of DM)	Casein infusion	83	37.6	6.11	21.9	49.6
	Feeding trial	94	37.3	6.25	25.7	50.3
Digestible flow (g/d)						
Arginine	Casein infusion	83	83.4	24.80	31.5	162.5
	Feeding trial	83	100.9	22.44	56.3	168.8
Histidine	Casein infusion	83	39.5	12.37	18.7	80.9
	Feeding trial	90	46.9	9.80	27.6	76.7
Isoleucine	Casein infusion	83	87.9	25.37	35.9	176.1
	Feeding trial	94	105.7	18.89	61.8	163.2
Leucine	Casein infusion	83	151.4	46.88	60.5	304.1
	Feeding trial	94	188.2	40.50	123.0	295.7
Lysine	Casein infusion	83	119.3	34.01	50.7	231.8
	Feeding trial	94	139.4	24.35	80.0	215.7
Methionine	Casein infusion	81	34.5	10.49	14.6	67.1
	Feeding trial	94	40.5	7.22	23.8	59.2
Phenylalanine	Casein infusion	83	89.6	25.67	36.4	174.7
	Feeding trial	94	108.6	21.33	62.4	171.8
Threonine	Casein infusion	83	86.2	22.22	41.0	157.7
	Feeding trial	94	106.3	18.04	61.5	158.2
Valine	Casein infusion	83	99.7	28.46	43.4	204.5
	Feeding trial	94	120.1	20.56	69.6	179.5
Branched-chain AA	Casein infusion	83	339.0	100.03	139.9	679.8
	Feeding trial	94	414.0	79.37	254.4	638.4

[1]All variables were calculated using NRC (2001); balance = supply minus requirements for maintenance and lactation; branched-chain AA = Ile + Leu + Val.

[2]Number of observations.

[3]The infusate was not included in the computation of the variable.

Table 4. Correlation coefficients between moderator variables in feeding trials[1]

Item	NE_L supply	NE_L concentration	MP balance	NE_L balance	Concentrate	CP	RUP	NDF	NFC	AADI[2]
MP supply	0.88***	$0.09^{0.39}$	0.79***	0.54***	0.33***	0.42***	0.61***	−0.43**	0.27**	0.96*** to 0.99***
NE_L supply		$0.16^{0.13}$	0.56***	0.63***	0.32***	$0.12^{0.25}$	0.38***	−0.47***	0.36***	0.84*** to 0.90***
NE_L concentration			0.21*	0.24*	0.34***	$0.11^{0.28}$	$0.14^{0.19}$	−0.51***	0.40***	$0.04^{0.69}$ to 0.21*
MP balance				0.69***	0.37***	0.46***	0.54***	−0.41***	0.24*	0.76*** to 0.82***
NE_L balance					0.44***	$-0.03^{0.79}$	$0.20^{0.06}$	−0.51***	0.45***	0.52*** to 0.62***
Concentrate						$0.04^{0.72}$	0.25*	−0.40***	0.38***	0.30** to 0.39***
CP							$0.19^{0.07}$	$0.07^{0.51}$	−0.37***	0.35*** to 0.43***
RUP								−0.43***	0.27**	0.51*** to 0.62***
NDF									−0.85***	−0.37*** to −0.59***
NFC										0.22* to 0.38***

[1]All variables were calculated using NRC (2001). Refer to Table 3 for unit of variables. Pearson's product-moment correlation; df = 92 (except 81 and 88 for Arg digestible flow and His digestible flow, respectively). Degree of significance = *$P \le 0.05$; **$P \le 0.01$; ***$P \le 0.001$; P-value in superscript if not significant.

[2]AADI = digestible flow of individual EAA.

RESULTS AND DISCUSSION

Features of the Data Set

The data set included 2 independent sets of studies (36 CN and 42 FT). By study design, the diets fed to control and casein-infused cows were the same in CN studies, and ΔMP supply (and ΔAADI) was achieved by the infusion of casein. In casein-infused cows, the contribution of MP from CN to total MP supply averaged 17 ± 6.5% (range = 6–40%; data not shown). In contrast, ΔMP supply was accomplished by varying dietary CP, RUP (% of CP), NDF, or NFC concentrations in FT studies (Supplemental Figure S1; https://doi.org/10.3168/jds.2018-15218). Therefore, the influence of moderators was tested only in FT studies because continuous moderators showed within-study variation (Table 7).

In FT studies, the meta-design was investigated for the independence of association among the levels of categorical moderators using the chi-squared test. The effect of geography was not tested as a categorical moderator because this factor is not driving responses per se; however, breed and forage were associated with geography in the current data set. All studies with Ayrshire cows (35 treatment means) were European studies, whereas most studies with Holstein cows were conducted in America (55 out of 59 treatment means; $P < 0.001$). All Ayrshire cows were fed grass- or legume-based diets and most Holstein cows were fed mixed forage diets (47 out of 59 treatment means; $P < 0.001$). Most dairy cows were in early lactation (29 Ayrshire and 36 Holstein cows; $P = 0.05$). Therefore, FT studies were not well balanced for breed, forage, and stage of lactation, and results related to these factors should be interpreted cautiously.

Casein Infusion Studies Versus Feeding Trials in the Subset of Data

The meta-design was not balanced for DMI and the supplies of MP and NE_L in the initial data set (Figure 1); therefore, a subset of data was created to balance the meta-design between CN and FT studies for the supplies of MP and NE_L. The subset of data set included 17 CN and 19 FT studies (35 and 49 treatment means, respectively). In the subset of data (1) DMI averaged 18.5 ± 1.17 and 20.0 ± 1.08 kg/d; (2) MP supply averaged 2.0 ± 0.33 and 1.9 ± 0.21 kg/d; and (3) NE_L supply averaged 29.7 ± 2.01 and 30.8 ± 1.18 Mcal/d in CN and FT studies, respectively.

The linear slope of each outcome was similar ($P \geq 0.19$) in CN and FT studies except for that of p[urea], which was twice lower ($P = 0.03$) in CN versus FT

Table 5. Comparison of linear terms between casein infusion studies (CN) and feeding trials (FT) in the subsetted data set[1]

	CN studies[3]			FT studies[3]			CN vs. FT
Outcome[2]	N	Q_E	Best equation[4]	N	Q_E	Best equation[4]	P-value
MTPY	36	507	314*** + (270*** × MP)	54	610	458*** + (231*** × MP)	0.70
Urea	26	3,307	$1.26^{0.45}$ + (1.65* × MP)	36	2,849	−2.50* + (3.41*** × MP)	0.03
Arg	36	175	$28^{0.21}$ + (0.49* × ArgDI)	54	470	$31^{0.10}$ + (0.57* × ArgDI)	0.74
His	36	408	$-23^{0.09}$ + (1.53*** × HisDI)	54	441	$-21^{0.25}$ + (1.46** × HisDI)	0.75
Ile	36	583	$-9.4^{0.49}$ + (1.40*** × IleDI)	52	391	$-3.2^{0.91}$ + (1.47*** × IleDI)	0.76
Leu	36	576	−61** + (1.25*** × LeuDI)	54	1,320	−57* + (1.24*** × LeuDI)	0.64
Lys	36	373	$-8.7^{0.51}$ + (0.72*** × LysDI)	54	240	$-0.4^{0.98}$ + (0.68** × LysDI)	0.98
Met	34	328	14*** + (0.23* × MetDI)	52	707	$9.1^{0.07}$ + (0.30* × MetDI)	0.65
Phe	36	118	26*** + (0.21*** × PheDI)	52	218	25*** + (0.22** × PheDI)	0.80
Thr	36	455	96*** + (0.18*** × ThrDI)	52	286	64** + (0.47* × ThrDI)	0.19
Val	36	915	−64* + (3.27*** × ValDI)	52	611	$-131^{0.09}$ + (3.69*** × ValDI)	0.78
BCAA	36	809	−143* + (1.83*** × BCAADI)	54	560	$-103^{0.37}$ + (1.75*** × BCAADI)	0.68

[1]A mixed-effects multilevel model was used to evaluate separately the linear relationships in CN and FT studies from the subsetted data set. Linear terms were compared allowing the residual heterogeneity to be different in each set of studies.

[2]MTPY = milk true protein yield (g/d); plasma concentrations of urea (mM) and AA (μM); BCAA = branched-chain AA (Ile + Leu + Val).

[3]N = number of observations; Q_E = test for the residual heterogeneity ($P < 0.001$ for all relationships); MP = total MP supply (kg/d; infusate included); DI = digestible flow of AA (g/d). All moderators were calculated using NRC (2001).

[4]Degree of significance = *$P \le 0.05$; **$P \le 0.01$; ***$P \le 0.001$; P-value in superscript if not significant.

studies, indicating a greater urea synthesis or a lower recycling to the gut in FT versus CN studies (Table 5). The agreement among data sets is an indirect indication that NRC (2001) accurately predicted AADI in FT studies, assuming that ΔAADI calculated with casein infusion is our best representation of true variation in supply.

Milk True Protein Yield and Plasma Urea Concentration in the Large Data Set

Milk true protein yield was related linearly and quadratically ($P \le 0.03$) to MP supply in CN and FT studies (Table 6; Figure 2). Only 2 CN studies with high MP supply were available; thus, more research under high feeding levels is required to substantiate the quadratic relationship in CN studies. The quadratic relationships are in line with previous research showing that the marginal efficiency of milk protein yield relative to MP supply decreases with increased supply (Doepel et al., 2004; Metcalf et al., 2008; Daniel et al., 2016). Several moderators were associated with the decreased marginal efficiency in converting increasing amounts of MP supply into MTPY in FT studies, specifically dairy cows fed corn silage-based diets, fed in excess of energy requirements, or fed diets rich in CP and RUP (Table 7).

The p[urea] was related linearly ($P < 0.001$) to MP supply in CN studies and linearly and quadratically ($P \le 0.04$) in FT studies (Table 6; Figure 2). Results should be interpreted cautiously because data on urea concentration were reported in 46 studies only. The residual heterogeneity was abnormally high for each model, and only 2 CN studies (Mackle et al., 1999) supplied 2.5 kg of MP supply per day or more (Figure 2). In line with results from Table 5, the response in p[urea] to MP supply was nearly 2 times higher in FT than in CN studies, and the response was influenced positively by RDP in FT studies (Tables 6 and 7). The supplies of MP and RDP were negatively correlated (Table 4), indicating that FT studies with high MP supply were associated with low RDP and vice versa. The supply of MP averaged, respectively, 1,826 and 2,086 g/d in FT studies with high and low RDP levels ($P < 0.001$; data not shown). This correlation between MP and RDP suggests that ruminal ammonia load and urea synthesis decreased as MP supply increased due to decreasing RDP levels in the diet.

Plasma Concentrations of Isoleucine, Leucine, and Valine in the Large Data Set

The response of p[Ile], p[Leu], p[Val], and p[BCAA] to their respective AADI shared a common pattern that distinguished them from the other AA (Table 6). In CN studies, the linear relationships were significant ($P < 0.001$) but the quadratic relationships were not ($P \ge 0.12$); in contrast, the linear and quadratic relationships were both significant ($P \le 0.03$) in FT studies (Table 6; Figure 3). The linear slopes were very consistent among CN studies but were not consistent among FT studies; however, linear slopes obtained from the subset of data were similar, indicating that the relationships between p[AA] and AADI were not influenced by the addition of CN studies with low MP supply (Table 5). Our results are similar to those reported in Patton et al. (2015)

Table 6. Average effect size estimates in casein infusion studies (CN) and feeding trials (FT) from the large data set[1]

	CN studies[3]			FT studies[3]		
Outcome[2]	N	Q_E	Best equation[4]	N	Q_E	Best equation[4]
MTPY	85	2,818	254*** + (261*** × MP)	102	1,660	586*** + (173*** × MP)
	85	2,206	112* + (429*** × MP) − (46*** × MPsqr)	102	1,658	235$^{0.16}$ + (497** × MP) − (72* × MPsqr)
Urea	52	7,521	1.31$^{0.09}$ + (1.73*** × MP)	60	5,712	−3.38** + (3.19*** × MP)
	52	7,495	1.36$^{0.28}$ + (1.7$^{0.13}$ × MP) − (0.01$^{0.95}$ × MPsqr)	60	5,562	−9.0** + (8.2** × MP) − (1.05* × MPsqr)
Arg	81	597	39** + (0.41* × ArgDI)	89	954	41*** + (0.40*** × ArgDI)
	81	559	−9.3$^{0.59}$ + (1.5*** × ArgDI) − (0.006*** × ArgDIsqr)	89	949	11$^{0.69}$ + (1.0$^{0.06}$ × ArgDI) − (0.003$^{0.18}$ × ArgDIsqr)
His	85	2,861	−6.0$^{0.58}$ + (1.18*** × HisDI)	96	1,277	−5.0$^{0.63}$ + (0.98*** × HisDI)
	85	2,315	−50* + (3.2*** × HisDI) − (0.022*** × HisDIsqr)	96	971	−68* + (3.7** × HisDI) − (0.028** × HisDIsqr)
Ile	83	1,122	3.8$^{0.70}$ + (1.27*** × IleDI)	98	1,557	53* + (0.65** × IleDI)
	83	1,086	−13$^{0.53}$ + (1.68*** × IleDI) − (0.002$^{0.30}$ × IleDIsqr)	98	1,519	−112$^{0.09}$ + (3.6** × IleDI) − (0.013* × IleDIsqr)
Leu	83	1,311	−42** + (1.12*** × LeuDI)	102	2,795	1.3$^{0.94}$ + (0.78*** × LeuDI)
	83	1,302	−70* + (1.53*** × LeuDI) − (0.001$^{0.21}$ × LeuDIsqr)	102	2,579	−139*** + (2.2*** × LeuDI) − (0.004** × LeuDIsqr)
Lys	85	1,225	12$^{0.37}$ + (0.58*** × LysDI)	102	775	35** + (0.32** × LysDI)
	85	1,180	−34$^{0.14}$ + (1.3*** × LysDI) − (0.003* × LysDIsqr)	102	729	−61$^{0.17}$ + (1.7** × LysDI) − (0.005* × LysDIsqr)
Met	83	956	13*** + (0.21*** × MetDI)	100	1,662	18*** + (0.06$^{0.45}$ × MetDI)
	83	876	7.4* + (0.5** × MetDI) − (0.004** × MetDIsqr)	100	1,643	−7.2$^{0.48}$ + (1.3* × MetDI) − (0.015* × MetDIsqr)
Phe	83	329	34*** + (0.13*** × PheDI)	100	1,017	33*** + (0.12** × PheDI)
	83	329	31*** + (0.20** × PheDI) − (<0.001$^{0.21}$ × PheDIsqr)	100	1,017	30$^{0.12}$ + (0.2$^{0.56}$ × PheDI) − (<0.001$^{0.84}$ × PheDIsqr)
Thr	81	1,458	96*** + (0.15* × ThrDI)	100	623	95*** + (0.06$^{0.50}$ × ThrDI)
	81	1,286	81*** + (0.5* × ThrDI) − (0.001* × ThrDIsqr)	100	623	1.1$^{0.98}$ + (1.8* × ThrDI) − (0.008* × ThrDIsqr)
Val	81	1,585	−81** + (3.21*** × ValDI)	100	1,696	46$^{0.38}$ + (1.68*** × ValDI)
	81	1,574	−122* + (4.1** × ValDI) − (0.004$^{0.51}$ × ValDIsqr)	100	1,604	−389* + (8.5** × ValDI) − (0.026** × ValDIsqr)
BCAA	83	1,434	−110* + (1.73*** × BCAADI)	102	1,505	114$^{0.14}$ + (0.97*** × BCAADI)
	83	1,402	−223** + (2.4*** × BCAADI) − (0.001$^{0.12}$ × BCAADIsqr)	102	1,442	−397$^{0.09}$ + (3.3** × BCAADI) − (0.003* × BCAADIsqr)

[1]A mixed-effects multilevel model was used to evaluate separately the linear (Lin) and quadratic (Quad) relationships in CN and FT studies from the large data set.

[2]MTPY = milk true protein yield (g/d); plasma concentrations of urea (m*M*) and AA (µ*M*); BCAA = branched-chain AA (Ile + Leu + Val).

[3]N = number of observations; Q_E = test for the residual heterogeneity ($P < 0.001$ for all relationships); MP = total MP supply (kg/d; infusate included); DI = digestible flow of AA (g/d); MPsqr = MP × MP; BCAADIsqr = BCAADI × BCAADI. All moderators were calculated using NRC (2001).

[4]Degree of significance = $*P \leq 0.05$; $**P \leq 0.01$; $***P \leq 0.001$; *P*-value in superscript if not significant.

except for the significant quadratic relationship for Leu in the current study. Indeed, linear slopes for p[AA] versus AADuo should be about 20% lower compared with p[AA] versus AADI, because AADI equals on average 0.8 × AADuo (Supplemental File S1; https://doi.org/10.3168/2018-15218).

The BCAA belong to group 2 AA, which are minimally removed by the liver in dairy cows (Hanigan, 2005; Lapierre et al., 2012); therefore, the relationship between p[BCAA] and BCAADI was expected to be only linear. The quadratic component, significant for p[BCAA] only in FT studies, indicated that postabsorptive metabolism negatively influenced the response of p[BCAA] to incremental BCAADI in these studies. Several moderators were associated with the positive but declining response of p[BCAA] to increased BCAADI at high BCAADI supply in FT studies. These moderators included Holstein cows, dairy cows fed corn silage-based diets, or cows fed diets with high NE_L supply and rich in concentrate or RUP (Table 7). Average BCAADI was higher in FT studies separated into high and low NE_L supply levels (486 vs. 367 g/d, respectively; $P < 0.001$) and concentrate levels (438 vs. 388 g/d, respectively; $P < 0.001$; data not shown). In line with data in Table 7, linear slopes between p[BCAA] (μM) and BCAADI (g/d) were higher in low versus high

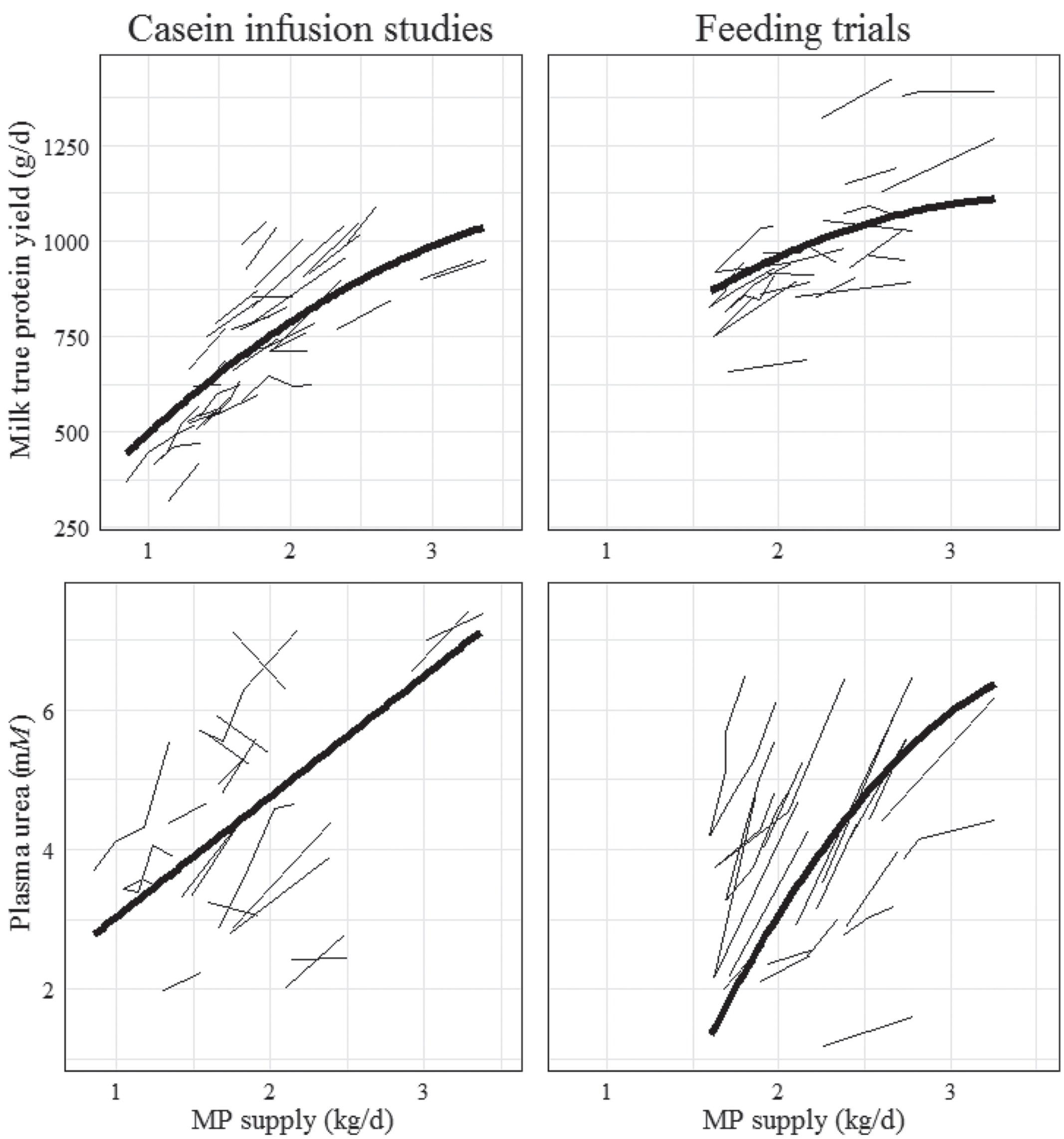

Figure 2. Relationships between milk true protein yield (g/d; upper panels) and plasma urea concentration (mM; bottom panels) versus MP supply (kg/d) in casein infusion studies and feeding trials (refer to Table 6). Each study is depicted by a connected line. The bold line represents the linear or quadratic relationship.

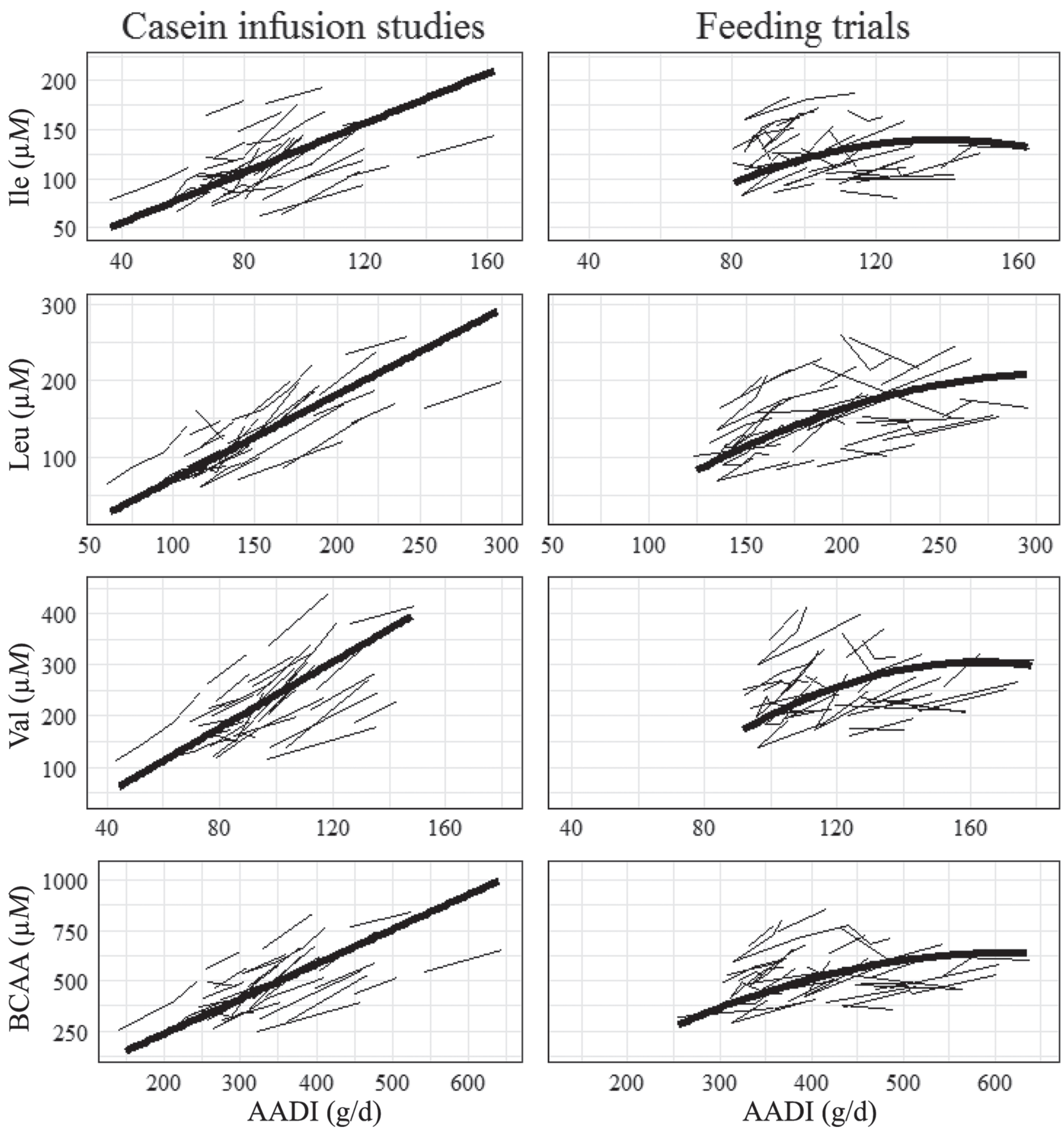

Figure 3. Relationships between plasma concentrations (μM) of Ile, Leu, Val, and branched-chain AA (BCAA) versus corresponding AA digestible flow (AADI; g/d) in casein infusion studies and feeding trials (refer to Table 6). Each study is depicted by a connected line. The bold line represents the linear or quadratic relationship.

levels of NE_L supply and concentrate. Linear slopes for low and high NE_L supply were, respectively (P-value in superscript), $1.22^{0.004}$ × BCAADI and $0.94^{<0.001}$ × BCAADI. Linear slopes for low and high concentrate levels were, respectively (P-value in superscript), $1.46^{<0.001}$ × BCAADI and $0.65^{0.01}$ × BCAADI (data not shown).

These observations are in agreement with studies that reported a decrease in p[BCAA] when dairy cows were fed starchy diets (Cantalapiedra-Hijar et al., 2014), when glucose was infused intravenously (Toerien et al., 2010; Curtis et al., 2014, 2018) or postruminally (Hurtaud et al., 1998; Lemosquet et al., 2009; Nichols et al., 2016), or when a gluconeogenic substrate was infused in the rumen (propionate; Raggio et al., 2006; Lemosquet et al., 2009) or postruminally (starch; Rius et al., 2010a). The suppressive effect of increased energy supply on p[BCAA] is associated with increased insulin secretion (Rius et al., 2010a; Nichols et al., 2016), an effect well demonstrated during the hyperinsulinemic-euglycemic clamp technique in dairy cows (Laarveld et al., 1981; Griinari et al., 1997; L'Espérance, 2000; Mackle et al., 2000) and other species (Castellino et al., 1990; Tesseraud et al., 1992; Bequette et al., 2001). During an insulin challenge, all p[EAA] decrease, but not as drastically as the p[BCAA], which can drop to

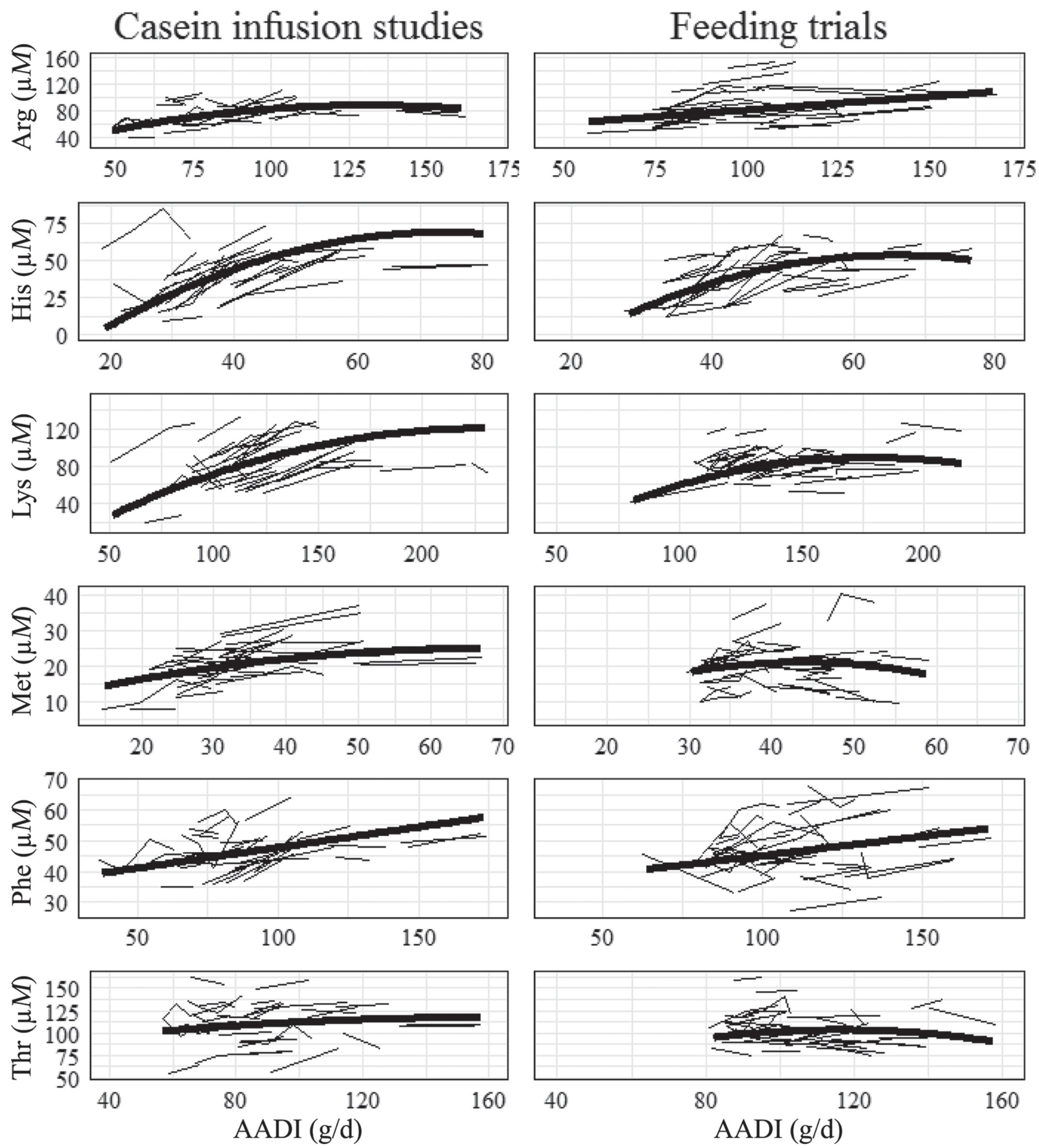

Figure 4. Relationships between plasma concentrations (μ*M*) of non-branched-chain amino acids versus corresponding AA digestible flow (AADI; g/d) in casein infusion studies and feeding trials (refer to Table 6). Each study is depicted by a connected line. The bold line represents the linear or quadratic relationship.

half control levels. Despite reviews showing the link between insulin, p[BCAA], glycemia, protein synthesis, and mammary function (e.g., mammalian target of rapamycin signaling pathway) over the last 3 decades (e.g., Lobley, 1992; Nair and Short, 2005; Zhang et al., 2017), more work is needed to elucidate the mechanisms regulating the insulinemic suppression of p[BCAA].

Plasma Concentrations of Other AA

Plasma concentrations of His, Lys, Met, and Thr were related quadratically ($P \leq 0.03$) to their respective AADI in CN and FT studies; p[Arg] was related ($P < 0.001$) quadratically to ArgDI in CN studies, but only linearly to ArgDI in FT studies; and p[Phe] was related linearly ($P < 0.01$) to PheDI in both sets of studies (Table 6; Figure 4). Although not identified as being influential observations using Cook's distance values, data from Mackle et al. (1999) appeared to influence the quadratic relationships in the visual appraisal of Figure 4. Models were retested forcing these potential outliers out from the model and the quadratic relationship for His was significant ($P = 0.04$) but the relationships for Arg, Lys and Thr were not ($P \geq 0.57$);

therefore, results for Arg, Lys, and Thr should be interpreted cautiously.

Overall, our results were similar to those reported by Patton et al. (2015) except that the quadratic relationship for His reached significance in the current study. The within-study slopes were less consistent for the non-BCAA, especially in FT studies (Figures 3 and 4). The same moderators previously identified for MTPY, p[urea], and group 2 AA were associated (in terms of direction and magnitude) with the declining response of plasma concentrations of His, Lys, Met, and Thr to their respective AADI in FT studies (Table 7).

Group 1 AA (His, Met, Phe, Trp, and Tyr) are largely removed by the liver (Hanigan, 2005; Lapierre et al., 2012). The hepatic removal appears to be mainly a mass action function of total liver inflow (Hanigan, 2005; Lapierre et al., 2005), and a higher removal by the liver would reduce the circulating concentrations and induce a significant quadratic term. In FT studies, the reduced response of p[His] to increased HisDI is in agreement with its inclusion in group 1 AA and with research indicating that liver extraction of His increases with His hepatic influx, largely related to His concentrations (Lapierre et al., 2014; Lapierre and Ouellet, 2015).

The quadratic relationship between p[Phe] and PheDI did not reach significance in Patton et al. (2015) and in the current study (Table 6; Figure 4). We can-

Table 7. Influence of moderators (B) on the linear relationship between each outcome and its moderator of interest (A) in feeding trials from the large data set[1]

Outcome[2]	A[3]	B[4]	N[5]	Q_E[6]	Best equation[7]
MTPY	MP supply	NE_L balance	102	904	$542^{***} + (208^{***} \times A)$ - $(0.2^{0.99} \times B) - (10^{*} \times A \times B)$
		CP	102	1,481	$-304^{0.22} + (589^{***} \times A) + (55^{***} \times B) - (26^{**} \times A \times B)$
		RUP	102	1,656	$61^{0.79} + (462^{***} \times A) + (13^{*} \times B) - (7.2^{*} \times A \times B)$
		Forage	102	1,496	$456^{***} + (238^{***} \times A) + (260^{*} \times B) - (121^{*} \times A \times B)$
Urea	MP supply	RDP	60	4,923	$6.4^{0.25} - (2.9^{0.21} \times A) - (0.2^{*} \times B) + (0.11^{**} \times A \times B)$
Arg	ArgDI	NE_L concentration	89	905	$-420^{*} + (3.9^{*} \times A) + (299^{*} \times B) - (2.3^{*} \times A \times B)$
		Breed	89	865	$8.2^{0.70} + (0.8^{**} \times A) + (49^{*} \times B) - (0.54^{*} \times A \times B)$
His	HisDI	NE_L supply	96	1,111	$-111^{0.07} + (3.8^{**} \times A) + (2.9^{0.12} \times B) - (0.07^{*} \times A \times B)$
		CP	96	980	$-112^{**} + (2.3^{**} \times A) + (8.3^{***} \times B) - (0.12^{*} \times A \times B)$
		RUP	96	950	$-79^{*} + (3.0^{***} \times A) + (1.8^{*} \times B) - (0.05^{**} \times A \times B)$
		Forage	96	1,010	$-28^{0.12} + (1.6^{***} \times A) + (45^{*} \times B) - (1.0^{*} \times A \times B)$
Lys	LysDI	NE_L supply	102	559	$-87^{0.16} + (1.6^{**} \times A) + (2.9^{0.09} \times B) - (0.03^{*} \times A \times B)$
		MP balance	102	775	$13^{0.41} + (0.5^{***} \times A) + (29^{0.20} \times B) - (0.29^{*} \times A \times B)$
		CP	102	672	$-149^{*} + (1.4^{**} \times A) + (13^{**} \times B) - (0.08^{**} \times A \times B)$
		Breed	102	671	$-19^{0.50} + (0.9^{***} \times A) + (67^{*} \times B) - (0.66^{**} \times A \times B)$
		Stage of lactation	102	749	$18^{0.29} + (0.4^{**} \times A) + (45^{*} \times B) - (0.34^{*} \times A \times B)$
Met	MetDI	NE_L supply	100	1,606	$-39^{*} + (1.2^{*} \times A) + (1.8^{**} \times B) - (0.03^{**} \times A \times B)$
		MP balance	100	1,558	$1.3^{0.81} + (0.5^{**} \times A) + (0.19^{0.97} \times B) - (0.22^{*} \times A \times B)$
		CP	100	1,588	$-21^{0.14} + (1.1^{**} \times A) + (2.2^{*} \times B) - (0.06^{*} \times A \times B)$
		RUP	100	1,343	$-14^{0.13} + (1.1^{***} \times A) + (0.7^{*} \times B) - (0.02^{***} \times A \times B)$
		Breed	100	1,638	$7.8^{0.11} + (0.3^{*} \times A) + (17^{**} \times B) - (0.42^{*} \times A \times B)$
Phe	PheDI	None	100	—	—
Thr	ThrDI	NE_L supply	100	608	$-93^{0.14} + (1.3^{*} \times A) + (5.8^{**} \times B) - (0.04^{*} \times A \times B)$
		MP balance	100	513	$35^{0.09} + (0.6^{**} \times A) + (16^{0.39} \times B) - (0.42^{**} \times A \times B)$
		CP	100	599	$-54^{0.26} + (1.2^{**} \times A) + (10^{**} \times B) - (0.08^{**} \times A \times B)$
		RUP	100	607	$-17^{0.67} + (1.5^{**} \times A) + (2.1^{*} \times B) - (0.03^{**} \times A \times B)$
		Breed	100	448	$51^{**} + (0.6^{**} \times A) + (53^{*} \times B) - (0.71^{**} \times A \times B)$
		Forage	100	490	$72^{***} + (0.3^{0.06} \times A) + (34^{0.08} \times B) - (0.37^{*} \times A \times B)$
BCAA	BCAADI	NE_L supply	102	1,207	$-600^{0.18} + (3.4^{**} \times A) + (18^{0.19} \times B) - (0.06^{*} \times A \times B)$
		Concentrate	102	1,505	$-977^{0.08} + (3.6^{**} \times A) + (23^{*} \times B) - (0.05^{*} \times A \times B)$
		RUP	102	1,432	$-587^{*} + (2.7^{***} \times A) + (19^{*} \times B) - (0.05^{*} \times A \times B)$
		Breed	102	1,364	$-247^{*} + (2.2^{***} \times A) + (431^{**} \times B) - (1.42^{***} \times A \times B)$
		Forage	102	1,125	$-141^{0.15} + (1.8^{***} \times A) + (331^{*} \times B) - (1.11^{**} \times A \times B)$

[1]A mixed-effects multilevel model was used to evaluate separately the influence of each moderator (B; see footnote 4) on the linear relationship between each outcome and its moderator of interest (A; see footnote 3). Only relationships with significant interaction term are reported.

[2]MTPY = milk true protein yield (g/d); plasma concentrations of urea (mM) and AA (μM); BCAA = branched-chain AA (Ile + Leu + Val).

[3]MP supply = total MP supply (kg/d; infusate included); DI = digestible flow of AA (g/d).

[4]NE_L balance (Mcal/d; infusate included); CP (% of diet DM); RUP (% of CP); forage (0 = grass/legume-based diets; 1 = corn silage-based diets); NE_L concentration (Mcal/kg of DM; infusate included); breed (0 = Ayrshire; 1 = Holstein); NE_L supply (Mcal/d; infusate included); stage of lactation (0 = early; 1 = mid-late); MP balance (kg/d; infusate included); concentrate = proportion of concentrate (%, DM basis). Continuous moderators were calculated using NRC (2001).

[5]N = number of observations.

[6]Q_E = test for the residual heterogeneity ($P < 0.001$ for all relationships).

[7]Degree of significance = $^{*}P \leq 0.05$; $^{**}P \leq 0.01$; $^{***}P \leq 0.001$; P-value in superscript if not significant.

not explain why Phe did not respond the same way the other group 1 AA did; however, p[Phe] was less consistent among studies compared with the other non-BCAA, especially at mid-PheDI (Figure 4). It is also interesting to note that Phe was the sole AA for which no moderator influenced the linear relationship in FT studies (Table 7).

The relationship between p[Lys] and LysDuo or LysDI was quadratic for feeding trials in Patton et al. (2015) and in the current study (Table 6; Figure 4). Hanigan (2005) reported that the hepatic removal of Lys and BCAA was proportionally less than the other AA; therefore, Lys belongs to group 2 AA and the relationships were expected to be similar to those observed for the BCAA. We cannot explain the quadratic relationships observed for Lys in this analysis, but several moderators influenced negatively the response of p[Lys] to increased LysDI in FT studies (Table 7).

Overall, our results from the FT database compared reasonably well with those reported by Patton et al. (2015). Casein infusion studies and feeding trials could not be compared in the large data set due to the lack of CN studies with high MP supply. Therefore, future research should investigate the effect of graded postruminal casein infusion on production and p[AA] responses in high-yielding dairy cows and explore further how these responses are altered also by energy supply.

CONCLUSIONS

The relationships between p[AA] and AADI in CN and FT studies were compared in a subset of data balanced for DMI and the supplies of MP and NE_L. The slopes were markedly similar for most individual AA in CN and FT studies, indicating indirectly that AADI predictions by NRC (2001) were accurate. In the data set that included all studies, p[BCAA] was related linearly to BCAADI in CN studies but quadratically to the other AA (except Arg and Phe) in FT studies. The within-study slopes were very consistent among CN studies for BCAA but more fluctuating for the non-BCAA. In FT studies, the response of p[BCAA] declined with increased BCAADI, and this effect was associated with diets supplying more energy, richer in concentrate, and based on corn silage. This decline in the response of p[BCAA] might be associated with the suppressive effect of insulin on BCAA.

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